GI Digestion and absorption

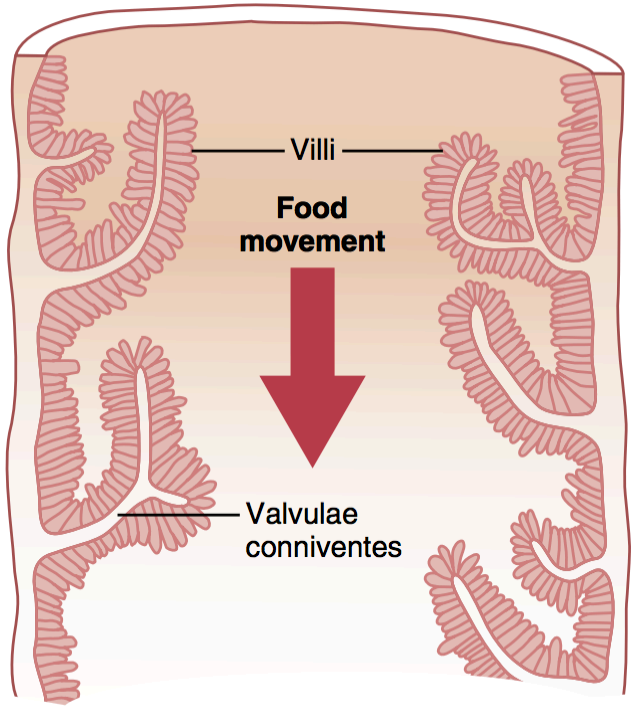
Pakkapon Rattanachaisit, MD

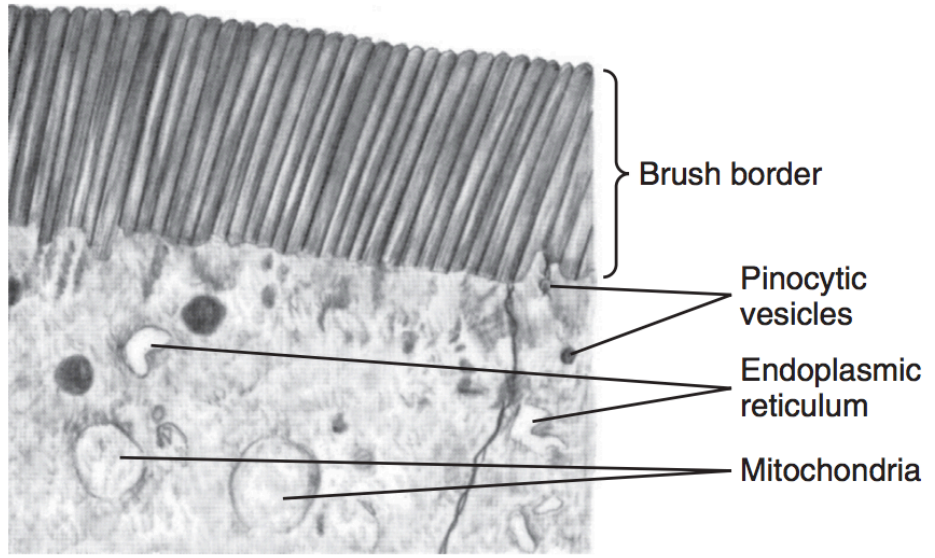
Outline

- Carbohydrate digestion and absorption
- Protein digestion and absorption
- Lipid digestion and absorption
- Vitamin absorption
- Mineral absorption
- Water and electrolyte absorption and secretion

Anatomical basis of absorption

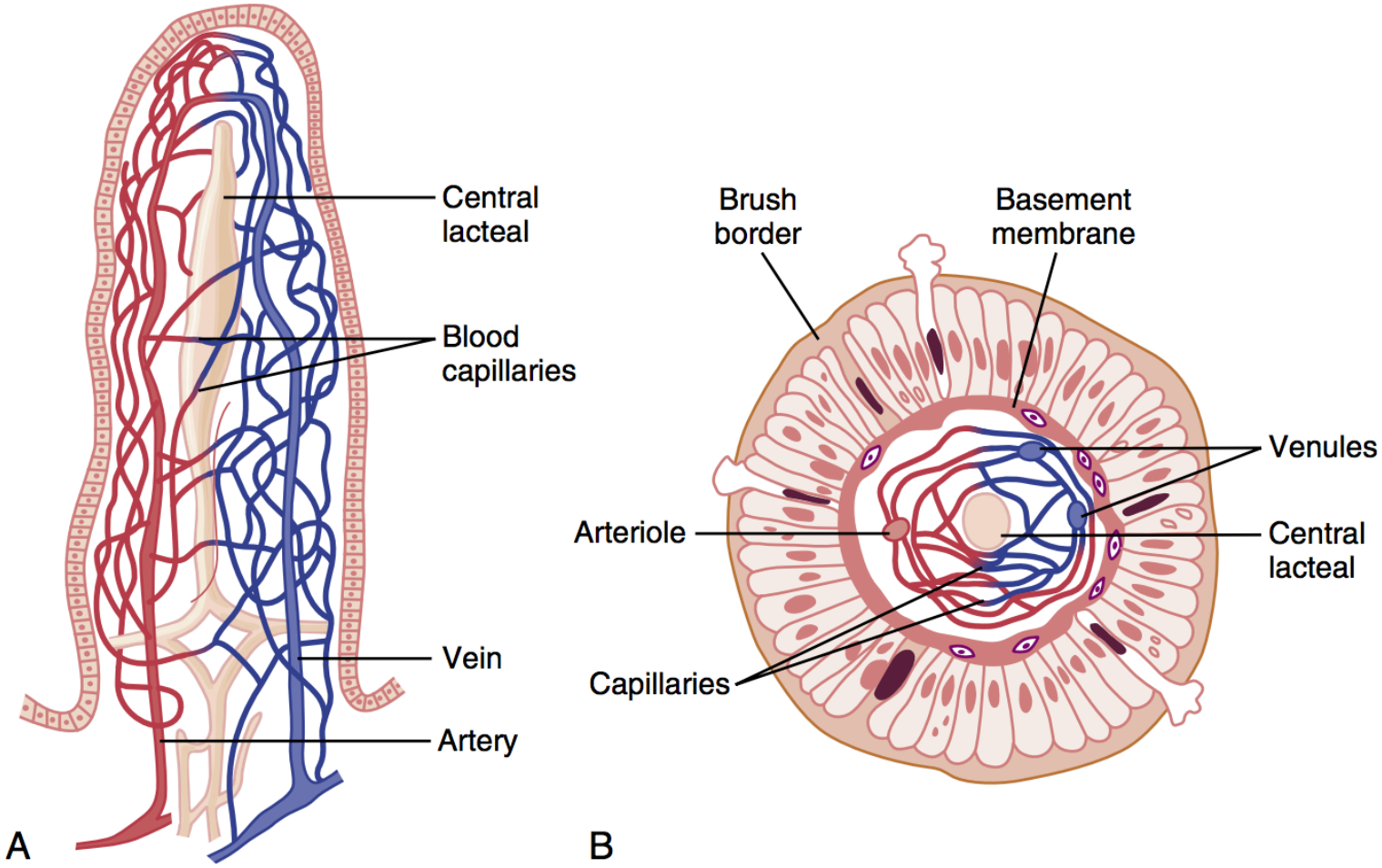
- Carbohydrates, proteins, and lipids are digested and absorbed in the small intestine.
- The surface area for absorption in the small intestine is greatly increased by the presence of the brush border





Guyton and Hall textbook of medical physiology. 14. ed., Elsevier; 2020.

The villus


Central lacteal
Blood capillaries
Vein
Artery
A
Brush border
Basement membrane
Venules
Arteriole
Central lacteal
Capillaries
B

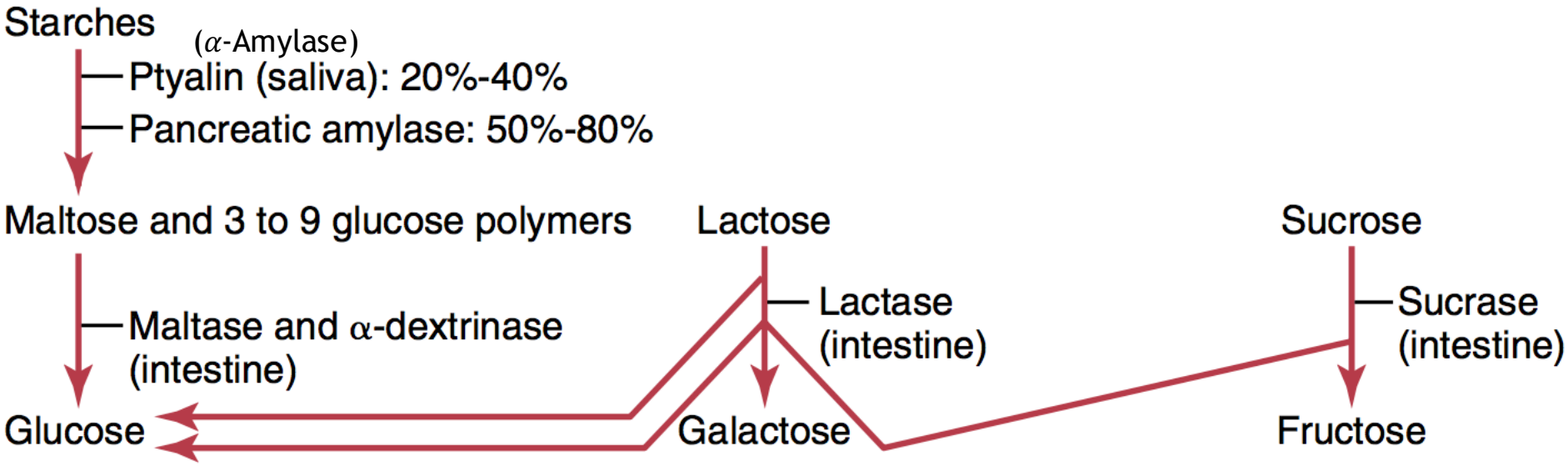
Carbohydrate digestion and absorption

- Three main forms of carbohydrate that have nutritional significance—starch, sucrose, and lactose.
- Other carbohydrates ingested to a slight extent are amylose, glycogen, alcohol, lactic acid, pyruvic acid, pectins, dextrins, and carbohydrate derivatives in meats.
- Only monosaccharides are absorbed. Carbohydrates must be digested to glucose, galactose, and fructose for absorption to proceed.
- Fiber cannot be degraded by mammalian digestive enzymes. It is critical for intestinal health: provides bulk to the stool, retains fluid and aids passage of the fecal material through the colon

Carbohydrate digestion

- The digestion of carbohydrates begins in the Mouth.
- Enzymes:
  - α-Amylases (salivary and pancreatic) hydrolyze 1,4-glycosidic bonds in starch, yielding maltose, maltotriose, and α-limit dextrins.
  - Maltase, α -dextrinase, and sucrase in the intestinal brush border then hydrolyze the oligosaccharides to glucose.
  - Lactase, trehalase, and sucrase degrade their respective disaccharides to monosaccharides.

Carbohydrate digestion

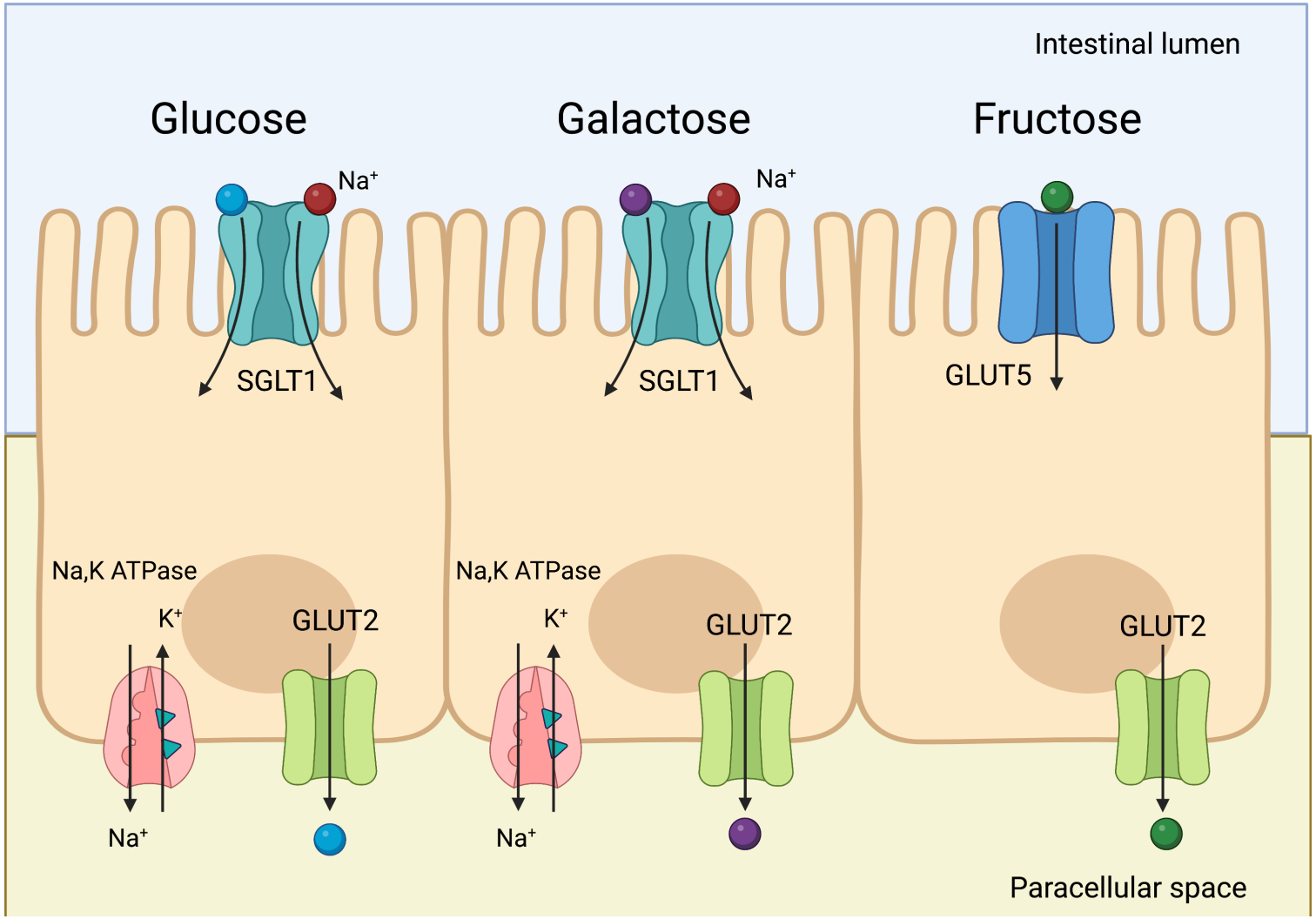


Brush border enzymes

- Maltase, α-dextrinase degrades maltose to glucose
- Lactase degrades lactose to glucose and galactose
- Sucrase degrades sucrose to glucose and fructose

Guyton and Hall textbook of medical physiology. 14. ed., Elsevier; 2020.

Carbohydrate absorption



Na^+-dependent cotransport (SGLT 1)

Facilitated diffusion (GLUT 5, GLUT 2)

After absorption > Capillary of small intestine > Portal vein

Protein digestion and absorption

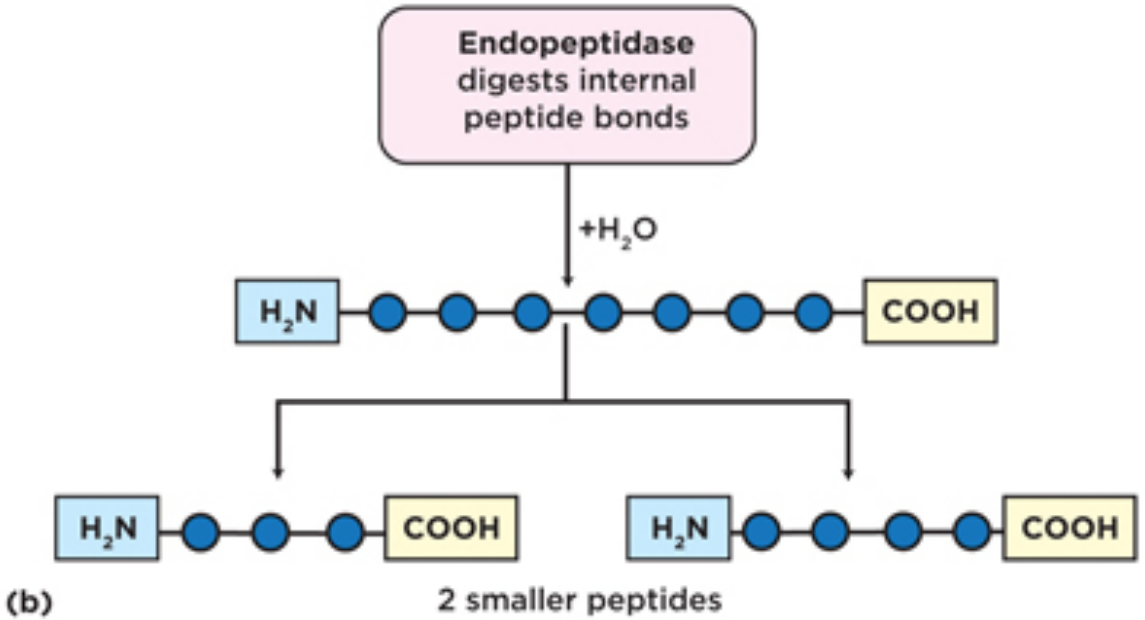
- Dietary proteins are chemically long chains of amino acids bound together by peptide linkages.
- There are 20 naturally occurring amino acids which require a broad spectrum of peptidases and transporters to mediate their digestion and uptake.
- Intestine is capable of transporting not only single amino acids, but also short oligomers, dipeptides, tripeptides, and even tetrapeptides.
- Thus, the final stage of protein digestion takes place in the cytosol of the enterocyte.

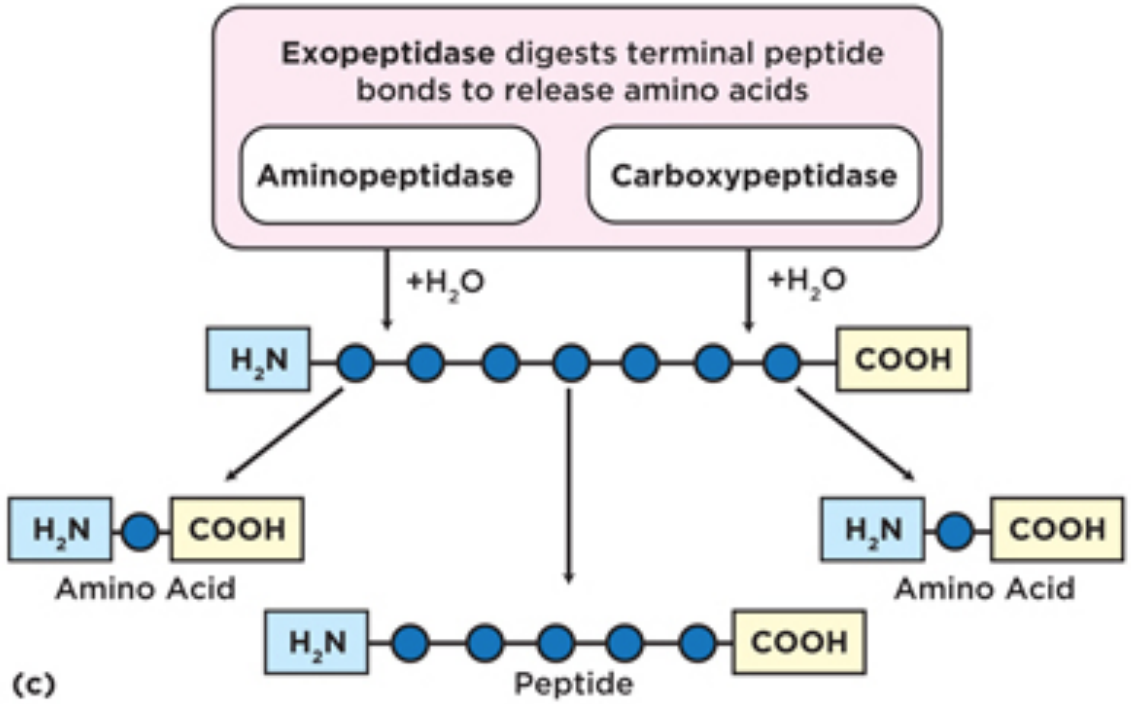
Gastric digestion of protein

- Digestion of dietary proteins begins in the stomach
- Pepsin
  - Pepsin secreted as pepsinogen by the chief cells of the stomach.
  - Pepsinogen is activated to pepsin by gastric H^+.
  - The optimum pH for pepsin is between 1 and 3.
  - When the pH is >5 (as in duodenum), pepsin is denatured.
- Gastric proteolysis results in incomplete digestion with only a few free amino acids; the products are mostly large, non-absorbable peptides
- Gastric proteolysis does not appear to be essential for normal levels of protein assimilation.

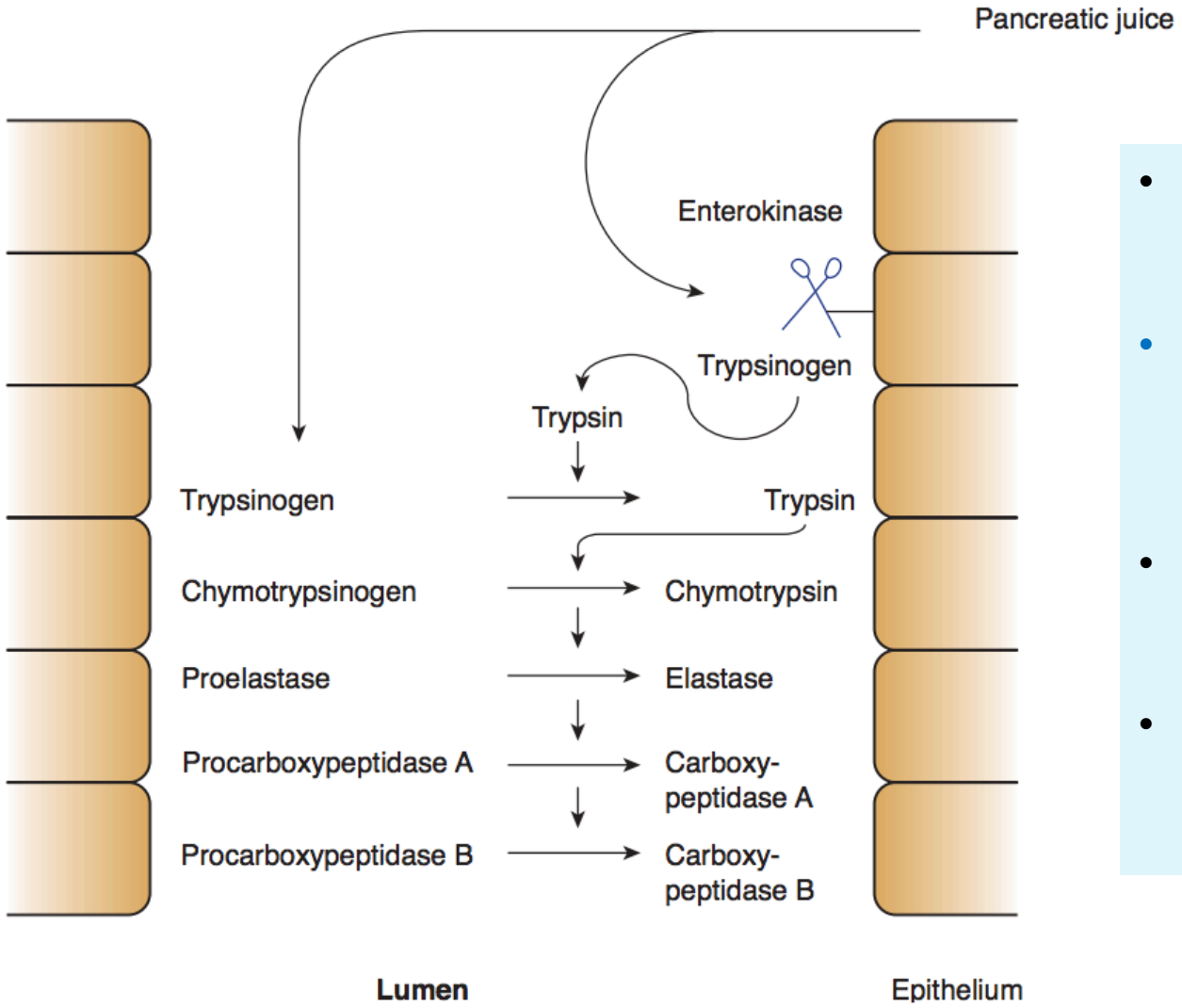
Intestinal digestion of protein

- Most protein digestion results from actions of pancreatic proteolytic enzymes.
- Two families of pancreatic proteases
  - Endopeptidases: degrade proteins by hydrolyzing interior peptide bonds.
  - Exopeptidases: hydrolyze one amino acid at a time from the C terminus of proteins and peptides.
- Pancreatic proteases
  - trypsin, chymotrypsin, elastase, carboxypeptidase A, and carboxypeptidase B
  - secreted in inactive forms that are activated in the small intestine


Endopeptidase digests internal peptide bonds
+H2O
H2N
COOH
H2N
COOH
H2N
COOH
(b)
2 smaller peptides


Exopeptidase digests terminal peptide bonds to release amino acids
Aminopeptidase
Carboxypeptidase
+H2O
+H2O
H2N
COOH
H2N
COOH
Amino Acid
H2N
COOH
Amino Acid
H2N
COOH
Peptide
(c)

Intestinal digestion of protein



- Mechanism to avoid activation of pancreatic proteases until they are in the duodenal lumen.
- Trypsinogen is cleaved by the enzyme enterokinase, which is expressed on the apical membrane of duodenal epithelial cells
- Even trypsinogen is converted to more trypsin by trypsin!
- After their digestive work is complete, the pancreatic proteases degrade each other and are absorbed along with dietary proteins.

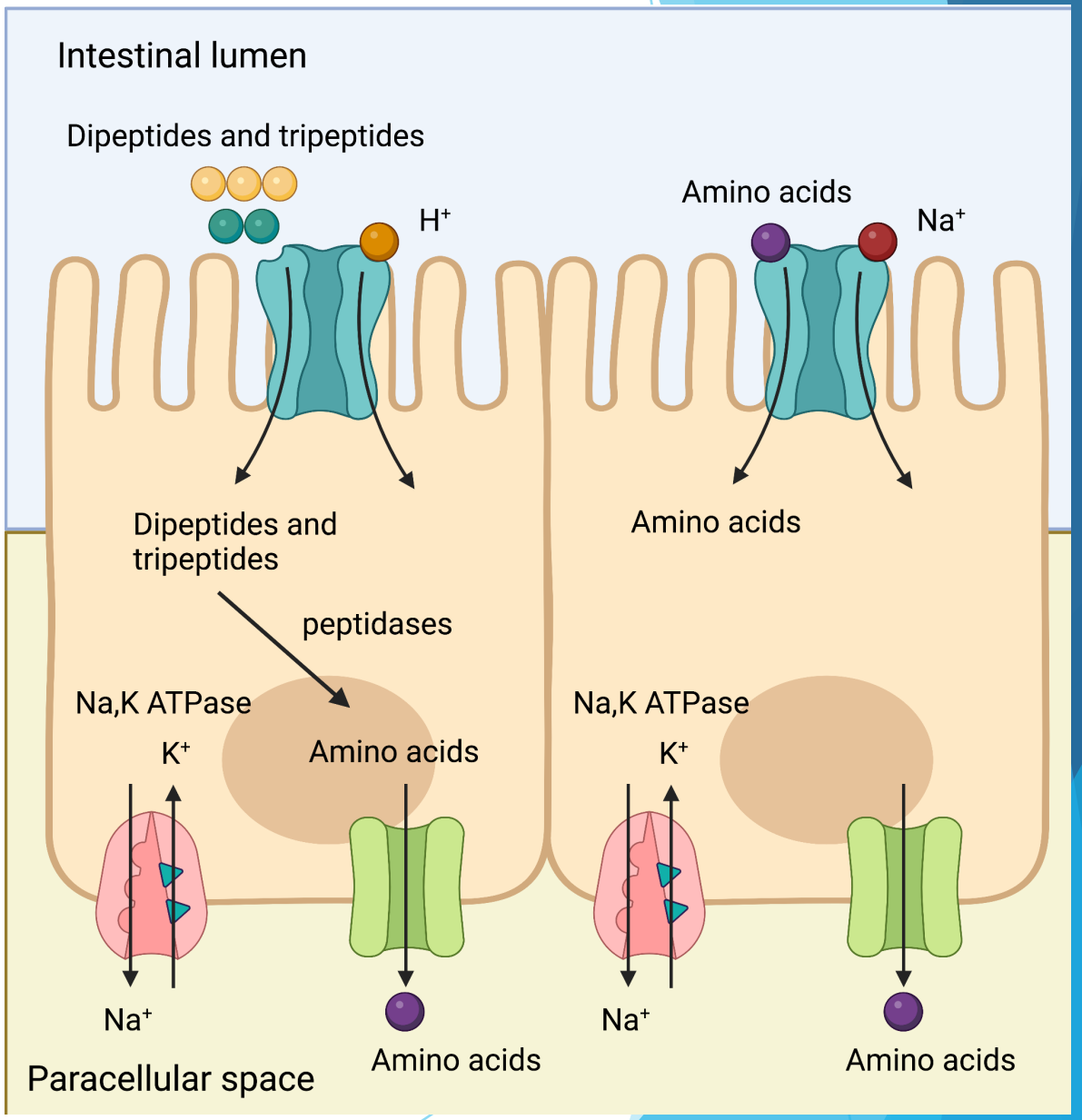
Ganong's Review of Medical Physiology, 23rd ed. McGraw-Hill Medical, 2009.

Intestinal digestion of protein

- Only small percentages of the proteins are digested all the way to their constituent amino acids by the pancreatic juices. Most remain as dipeptides and tripeptides.
- Brush border peptidases: split the remaining larger polypeptides into tripeptides and dipeptides and a few into amino acids.
- Amino acids, dipeptides, and tripeptides are transported through the microvillar membrane to the interior of the enterocyte.
- In the cystosol, all the last dipeptides and tripeptides are digested to the final stage to form single amino acids.

Absorption of proteins

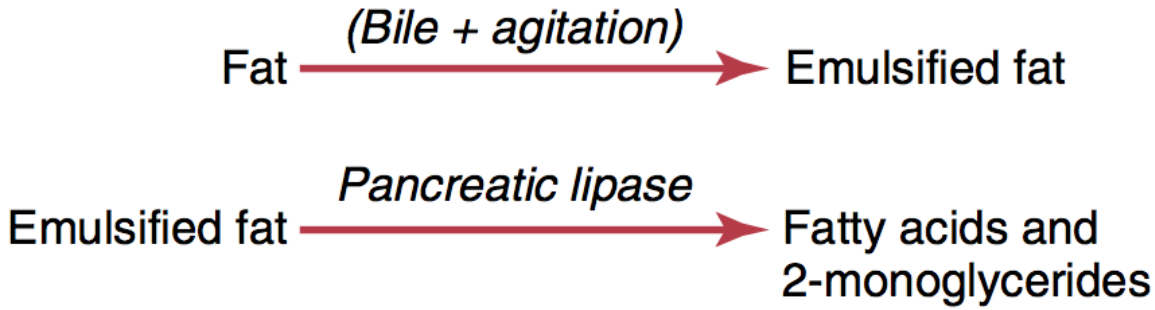
- Digestive products of protein can be absorbed as amino acids, dipeptides, and tripeptides
  - Na^+-dependent amino acid cotransport
  - H^+-dependent cotransport of dipeptides and tripeptides
- Dipeptides, and tripeptides are absorbed faster than free amino acids.
- Multiple transport systems for neutral, cationic, and anionic amino acids.



After absorption > Capillary of small intestine > Portal vein

Lipid digestion and absorption

- Lipids are hydrophobic and are more soluble in organic solvents (or cell membranes).
- The majority of lipid in the diet is in the form of triglycerides (each molecule of which is composed of a glycerol nucleus and three fatty acid side chains)
- Small quantities of phospholipids, cholesterol, and cholesterol esters are also present in the usual diet.
- Emulsification and Lipolysis



Lipid digestion: Stomach

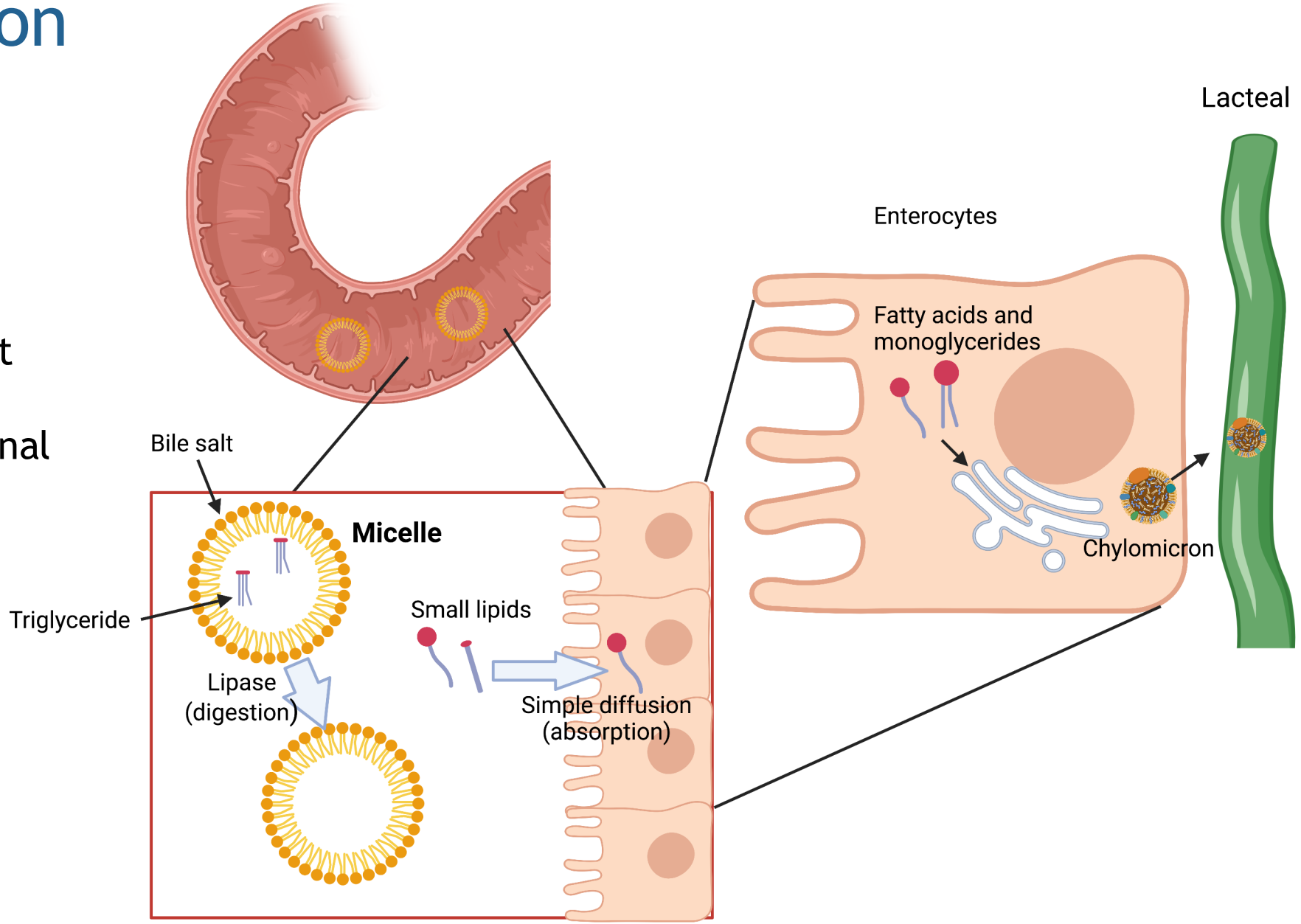
- Mixing breaks lipids into droplets to increase the surface area for digestion by pancreatic enzymes.
- Lingual lipases digest some of the ingested triglycerides to monoglycerides and fatty acids. However, most of the ingested lipids (>90%) are digested in the intestine by pancreatic lipases.
- CCK slows gastric emptying. Thus, delivery of lipids from the stomach to the duodenum is slowed to allow adequate time for digestion and absorption in the intestine.

Lipid digestion: intestine

- Fat digestion mainly occurs in small intestine
- Luminal events include;
  - Dispersion of the lipid into an emulsion by bile acids, thereby maximizing the area of the lipolysis
  - Lipolysis: pancreatic lipases hydrolyze lipids to fatty acids, monoglycerides, cholesterol, and lysolecithin.
  - The enzymes are pancreatic lipase, cholesterol ester hydrolase, and phospholipase A2.
  - Uptake of the products of lipolysis into **micelles** (aid by bile acids) that can then transfer these molecules to the epithelial surface

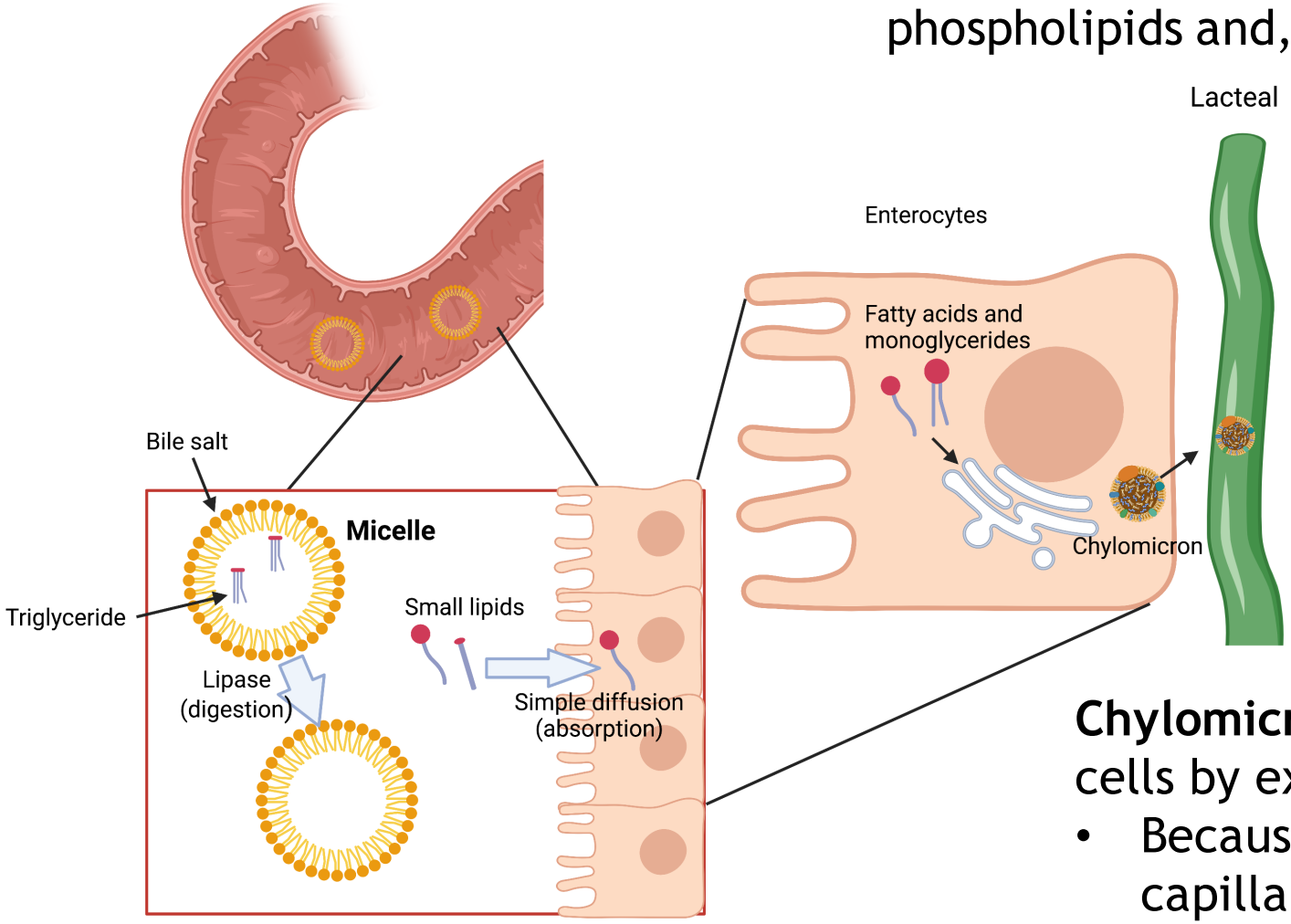
Lipid absorption

- Micelles bring the products of lipid digestion into contact with the absorptive surface of the intestinal cells.
- Then, fatty acids, monoglycerides, and cholesterol diffuse across the luminal membrane into the cells.



Lipid absorption

In the intestinal cells, the products of lipid digestion are **re-esterified** to triglycerides, cholesterol ester, and phospholipids and, with apoproteins, form **chylomicrons.**



Chylomicrons are transported out of the intestinal cells by exocytosis.

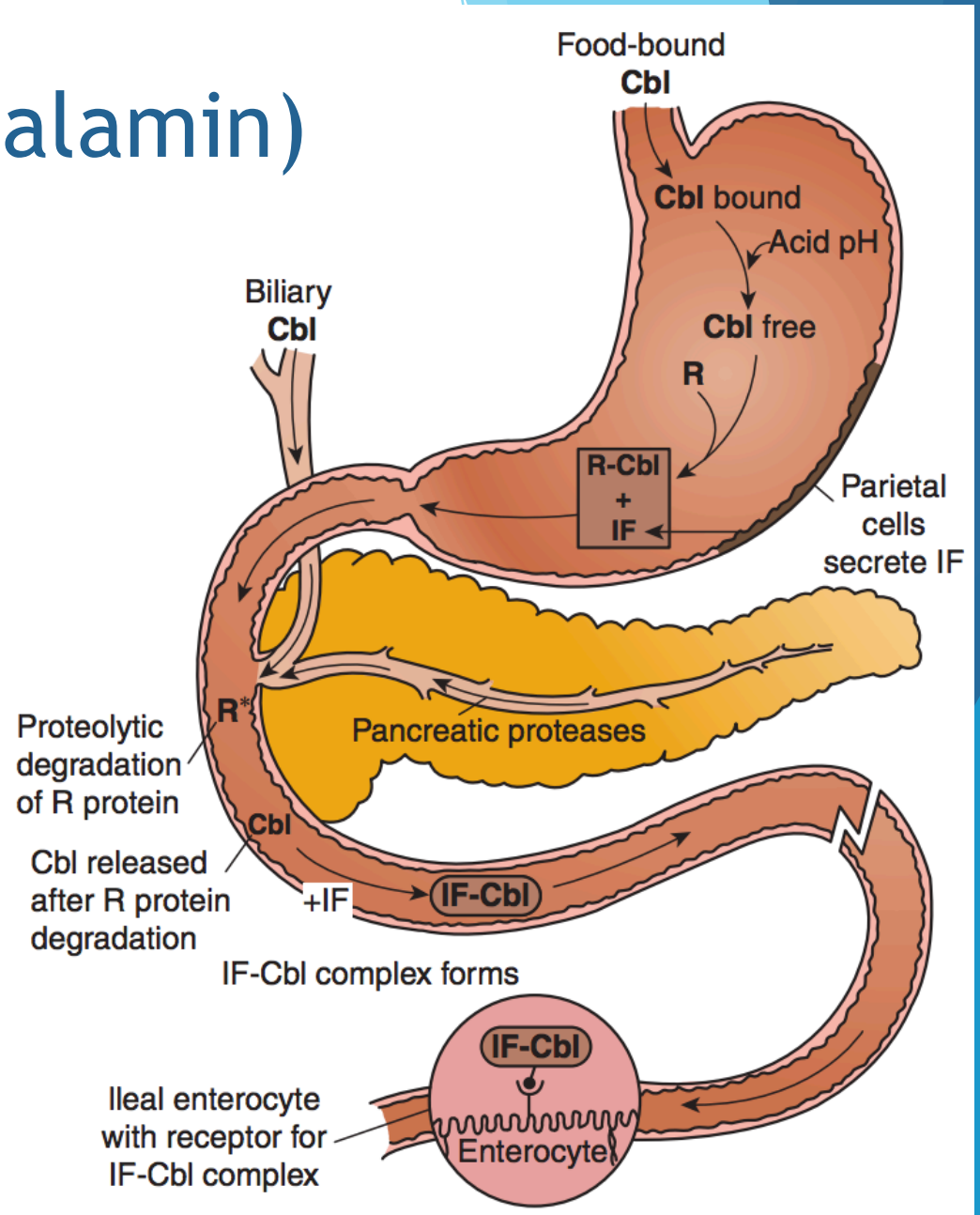
- Because chylomicrons are too large to enter the capillaries, they are transferred to lymph vessels > thoracic duct > bloodstream

Absorption of vitamins

- Vitamins are molecules that cannot be synthesized by the body, but which are essential to normal metabolism (act as cofactors for specific biochemical reactions)
- Most water-soluble vitamins are absorbed by Na^+-dependent cotransport mechanisms
  - thiamin, riboflavin, niacin, folate, vitamin B and vitamin C
- Fat-soluble vitamins (A, D, E, and K) are incorporated into micelles and absorbed along with other lipids.

Absorption of vitamin B12 (cobalamin)

- B12 is released from dietary proteins and binds initially to the R-binding protein (AKA haptocorrin) from salivary glands.
- Intrinsic factor cannot bind to cobalamin at low pH.
- In duodenum, R-binding protein is degraded by pancreatic proteases and B12 is transferred to intrinsic factor at the increased pH.
- In terminal ileum, complex of intrinsic factor and B12 binds to a receptor.
- Intrinsic factor is degraded, and the released B12 is then bound to plasma transcobalamin II > enter the circulation.



Textbook of Gastroenterology, 4th ed. Yamada T, et al. (editors). Philadelphia: Lippincott Williams and Wilkins, 2003.

Absorption of minerals

- Calcium
  - Actively absorbed into the blood, especially from the duodenum
  - Parathyroid hormone activates vitamin D, and the activated vitamin D in turn greatly enhances calcium absorption.
- Iron
  - Actively absorbed from the small intestine
  - Absorbed as heme iron (iron bound to hemoglobin or myoglobin) or as free Fe^{2+}.
  - In the intestinal cells, “heme iron” is degraded and free Fe^{2+} is released. The free Fe^{2+} binds to apoferritin and is transported into the blood.
  - Free Fe^{2+} circulates in the blood bound to transferrin, which transports it from the small intestine to its storage sites in the liver, and from the liver to the bone marrow for the synthesis of hemoglobin.

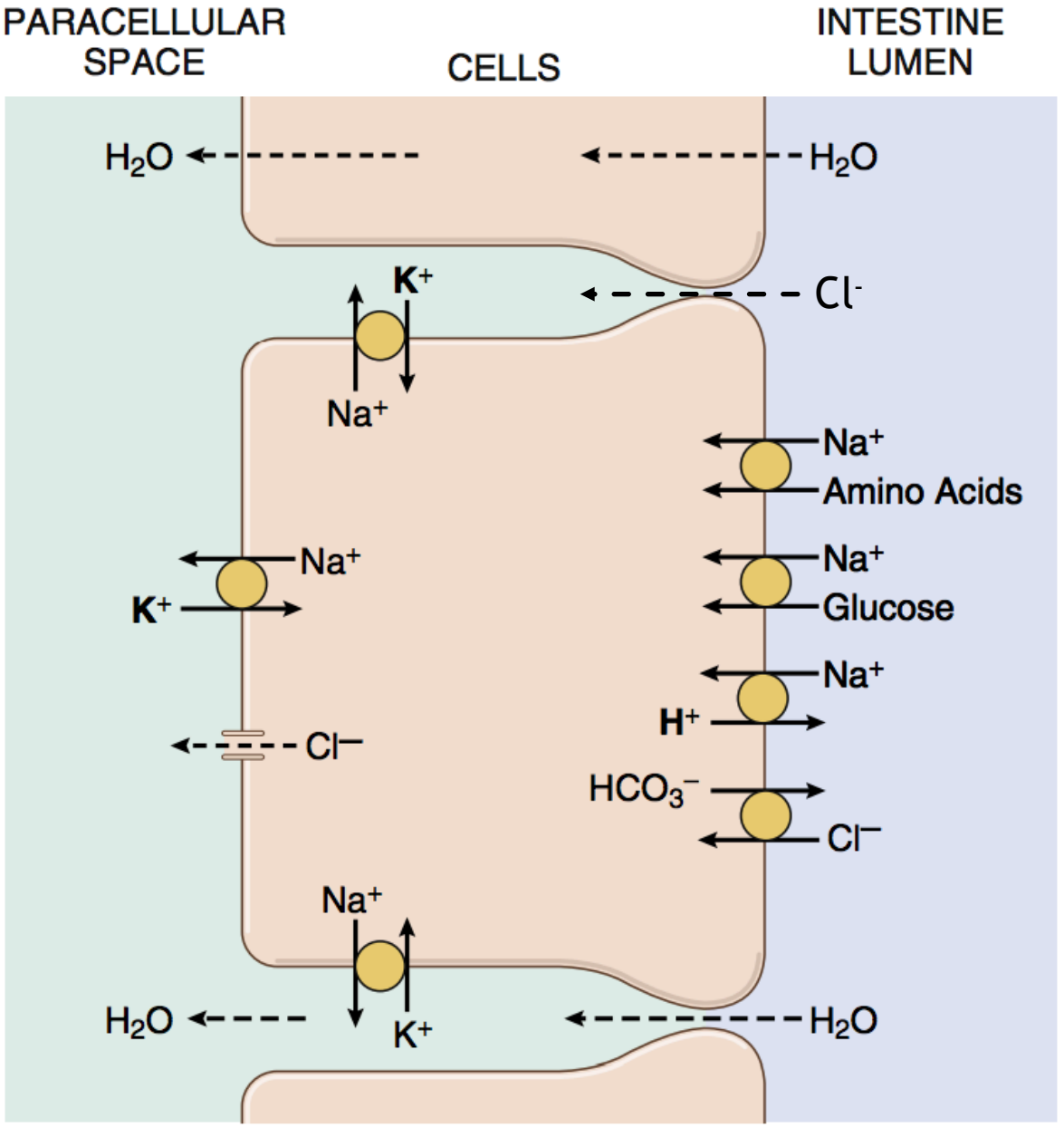
Absorption and secretion of water and electrolyte

- Electrolytes and H_2O may cross intestinal epithelial cells by either cellular or paracellular (between cells) routes.
- Tight junctions attach the epithelial cells to one another at the luminal membrane.
- The permeability of the tight junctions varies with the type of epithelium. A “tight” (impermeable) epithelium is the colon. “Leaky” (permeable) epithelia are the small intestine and gallbladder

Na^+ absorption in small intestine

- Na^+ moves into the intestinal cells, across the luminal membrane, and down its electrochemical gradient by the following mechanisms:
  (1) Passive diffusion (through Na^+ channels)
  (2) Na^+–glucose or Na^+–amino acid cotransport
  (3) Na^+–H^+ exchange

- Na^+ is pumped out of the cell against its electrochemical gradient by the Na^+–K^+ pump in the basolateral membranes.

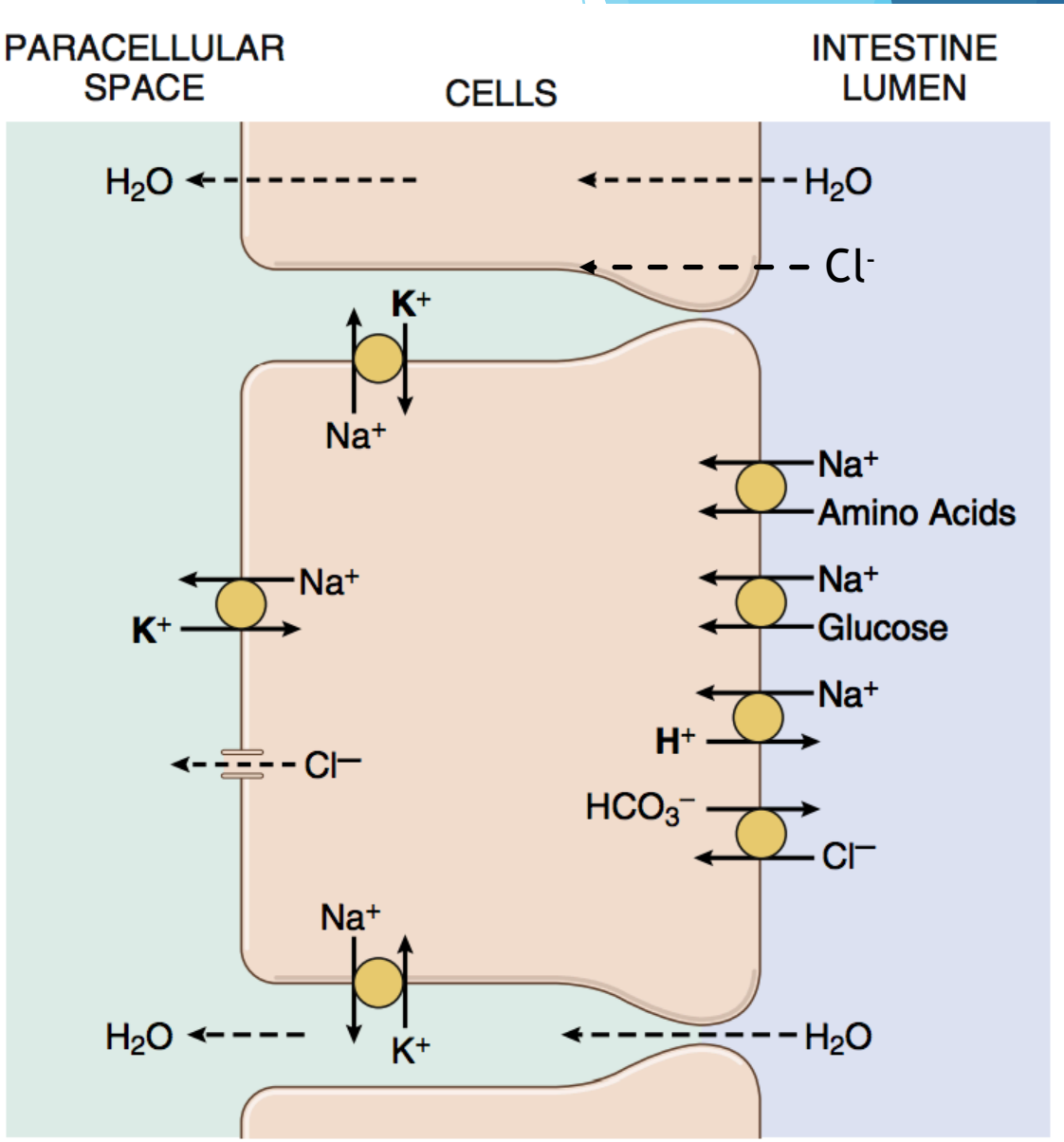


Guyton and Hall textbook of medical physiology. 14. ed., Elsevier; 2020.

Cl^- and H_2O absorption in small intestine

- Cl^- absorption accompanies Na^+ absorption throughout the GI tract :
  (1) Passive diffusion by a paracellular route
  (2) Na^+–Cl^- cotransport
  (3) Cl^-–HCO_3^- exchange

- Absorption of H_2O is secondary to solute absorption.



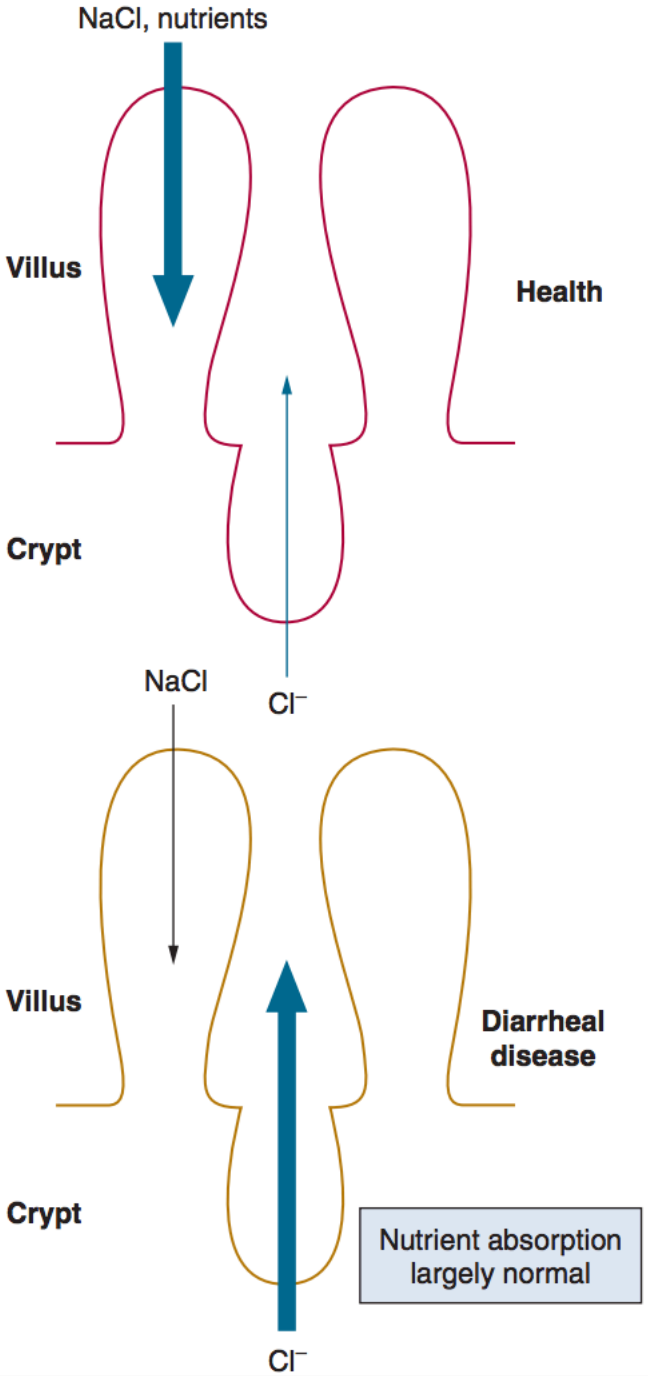
Guyton and Hall textbook of medical physiology. 14. ed., Elsevier; 2020.

Secretion of electrolytes and H_2O by the intestine

- The secretory mechanisms are located in the crypts. The absorptive mechanisms are located in the villi.

- Cl^- is the primary ion secreted into the intestinal lumen. It is transported through Cl^- channels in the luminal membrane that are regulated by cAMP.

- Na^+ is secreted into the lumen by passively following Cl^-. H_2O follows NaCl to maintain isosmotic conditions.



Balance between absorption and secretion in health and in secretory diarrheal disease. ➜

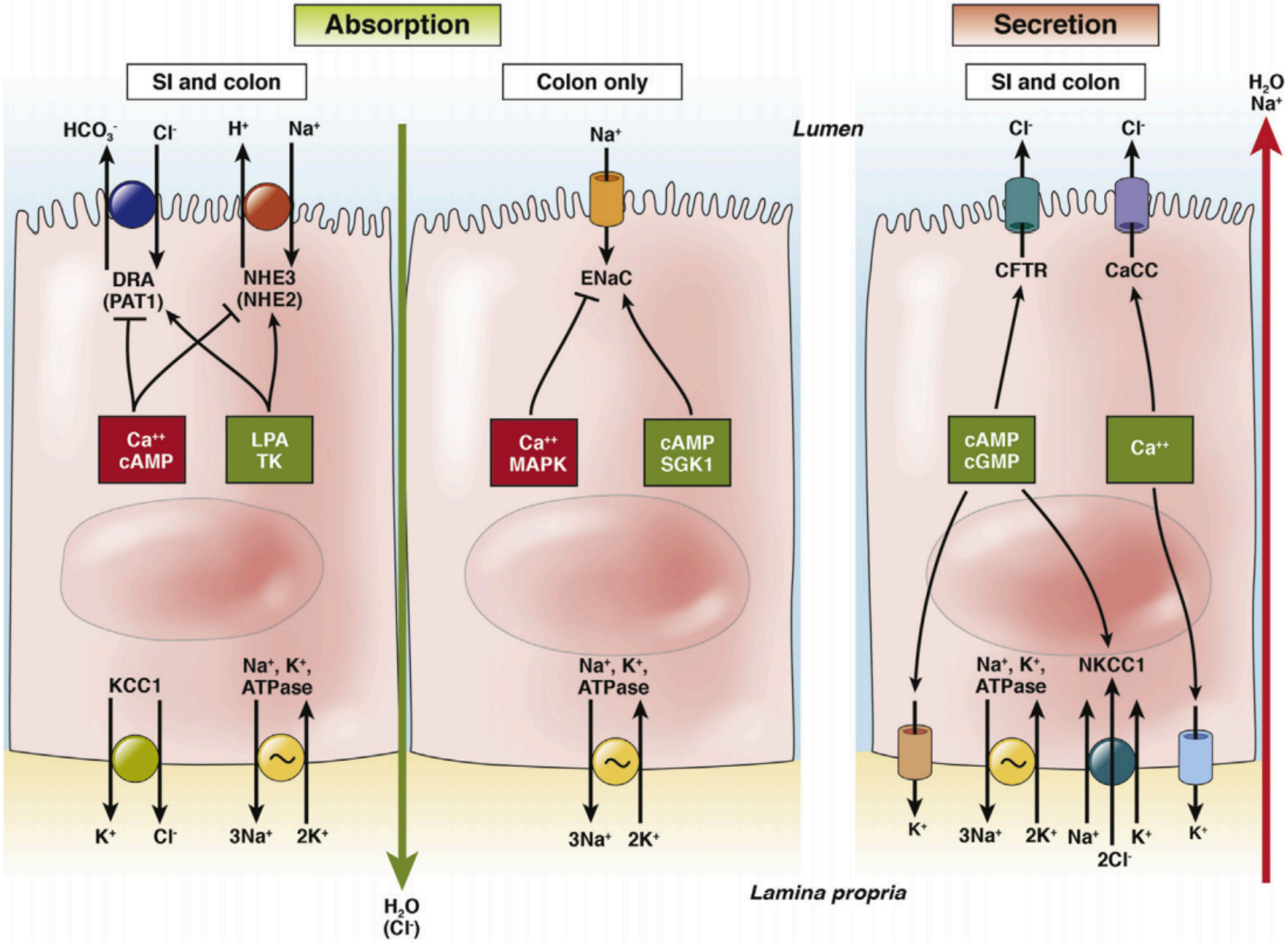


Figure 1. Cellular mechanisms accounting for intestinal absorption and secretion. Factors that reduce the amount or function of a specific transporter are shown in *red boxes*; those that increase levels of activity are shown in *green boxes*. *Long vertical arrows* indicate the paracellular absorption or secretion of water, with or without an appropriate counterion. LPA, lysophosphatidic acid; SI, small intestine; TK, tyrosine kinase.

Gastroenterology 2017;152:515-53

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